

DATA SCIENCE PROJECT REPORT

Matrix Mathematics for Artificial Neural Networks

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Executive Summary

This project introduces the matrix mathematics used in Artificial Neural Networks (ANNs). It demonstrates matrix creation, matrix multiplication, vector operations, and the role of linear algebra in forward propagation.

Concept	Purpose
Matrices	Store input features, weights, and biases
Matrix Multiplication	Compute weighted sums efficiently
Vectors	Represent inputs, outputs, and biases
Linear Algebra	Foundation of ANN computations

Introduction

Artificial Neural Networks rely heavily on matrix operations because they allow millions of computations to be performed efficiently during training and prediction.

Code Explanation

The notebook demonstrates creating matrices using NumPy, performing addition, subtraction, multiplication, transpose operations, and dot products. These operations simulate how neural networks combine inputs with weights before applying activation functions.

Importance in ANN

Every hidden layer computes outputs using matrix multiplication followed by activation functions. Matrix operations enable fast computation on CPUs and GPUs.

Applications

Matrix mathematics is used in deep learning, computer vision, natural language processing, recommender systems, robotics, and scientific computing.

Future Scope

Extend the notebook by implementing forward propagation, backpropagation, gradient descent, and full neural network training using NumPy or TensorFlow.

Conclusion

Understanding matrix mathematics provides the mathematical foundation required to study and build Artificial Neural Networks and modern deep learning models.